

EXECUTIVE INSIGHTS REPORT

OW Yellow Caps

NYC Yellow Taxi Operations | Q1 2025 (January – March)

Prepared for:

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Prepared by:

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Based on NYC Taxi & Limousine Commission (TLC) trip-record data, Jan–Mar 2025.

1. At a Glance — Q1 2025 KPI Dashboard

The table below summarises the eight most important performance indicators for OW Yellow Caps in the first quarter of 2025. These numbers are derived from 10.4 million validated trips recorded by the NYC TLC.

□ Source: [NB1](#) → *\$4 Audit Summary — Cell [24]* | [NB2](#) → *KPI Card — Cell [23]*

10.4M Total Trips <i>Jan–Mar 2025</i>	\$281.5M Total Revenue <i>All payment types</i>	\$18.54 Avg Fare / Trip <i>Excl. surcharges</i>	\$27.27 Avg Total / Trip <i>Incl. all fees</i>
15.6 min Avg Duration <i>Median: 12.1 min</i>	3.23 mi Avg Distance <i>Median: 1.76 mi</i>	70.4% Credit Card Share <i>of all trips</i>	+17.8% Q-o-Q Growth <i>Jan vs Mar trips</i>

March was the strongest month, generating \$105.2M in revenue — 21% more than January or February individually. This growth reflects both seasonal demand recovery and a potential post-CBD stabilisation effect.

Month	Total Trips	Revenue (\$M)	Avg Total (\$)
January 2025	3,264,227	\$86.5M	\$26.51
February 2025	3,318,704	\$86.7M	\$26.11
March 2025	3,846,652	\$105.2M	\$27.35

Table 1. Monthly trip volume and revenue summary, Q1 2025.

2. Demand Trends

2.1 Overall Trip Volume

Total trip volume grew steadily across the quarter. The daily average rose from approximately 105,000 trips per day in January to 124,000 in March. The chart below shows both daily fluctuations and a 7-day rolling average to remove the weekend/weekday cycle.

□ Source: [NB2](#) → Fig. 2 · 7-Day Rolling Average — Cell [4]

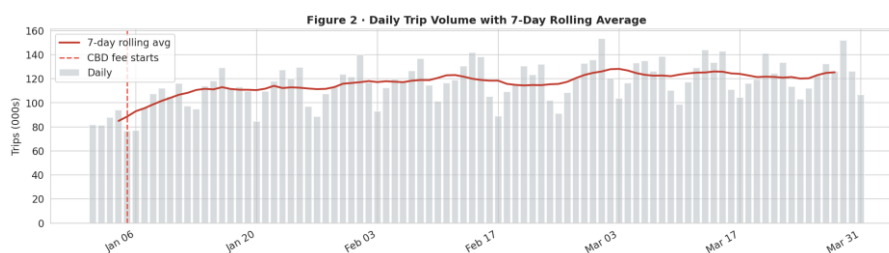


Figure 1. Daily trip volume (bars) and 7-day rolling average (red line), Jan–Mar 2025. The dashed vertical line marks the start of CBD congestion pricing on 5 January 2025.

2.2 Monthly Breakdown

The chart below shows daily and monthly volumes together. February shows fewer trips than January despite having similar days, partly due to winter weather effects. March's stronger volumes suggest pent-up demand and improved service utilisation.

□ Source: [NB2](#) → Fig. 1A · Daily Volume, Fig. 1B · Monthly Volume — Cell [3]

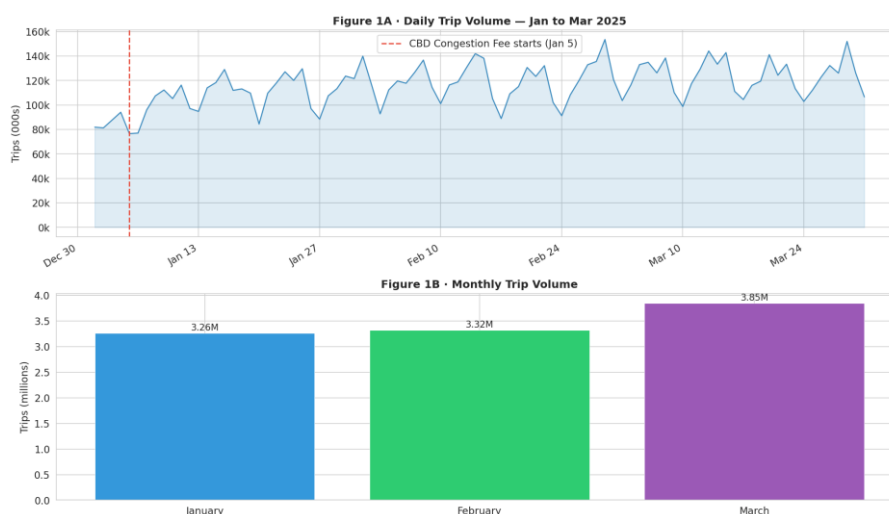


Figure 2. Daily trip volume (top) and total trips by month (bottom), Q1 2025.

3. When Customers Ride

3.1 Time of Day

Demand is highly concentrated in the afternoon and evening. The 6 PM hour is the single busiest of the day, with 742,000 trips — 10.6 times more than the quietest hour (4 AM). There are two clear peaks:

□ Source: [NB3](#) → Fig. 1 · Hourly Trip Demand — Cell [3]

- Morning commute peak: 8–10 AM (office workers, airport connections)
- Evening peak: 5–8 PM — the dominant peak, driven by end-of-work journeys, dining, and entertainment

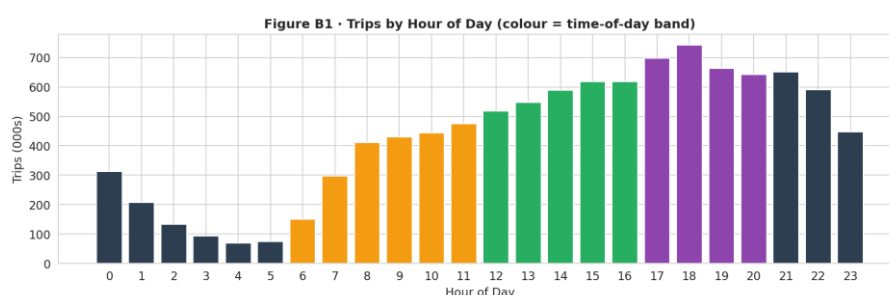


Figure 3. Total trips by hour of day, Jan–Mar 2025. Colours indicate time-of-day band: Night (dark), Morning (amber), Afternoon (green), Evening (purple).

3.2 Day of Week

The heatmap below reveals that the weekday evening peak (Thursday–Friday, 5–8 PM) is the most active period across the entire quarter. Saturday nights (9 PM–2 AM) are notably strong relative to other weekend hours, reflecting leisure and entertainment demand.

□ Source: [NB3](#) → Fig. 10 · Weekday × Hour Heatmap — Cell [19] | [NB3](#) → Fig. 2A/2B · Weekday Demand Profile — Cell [5]

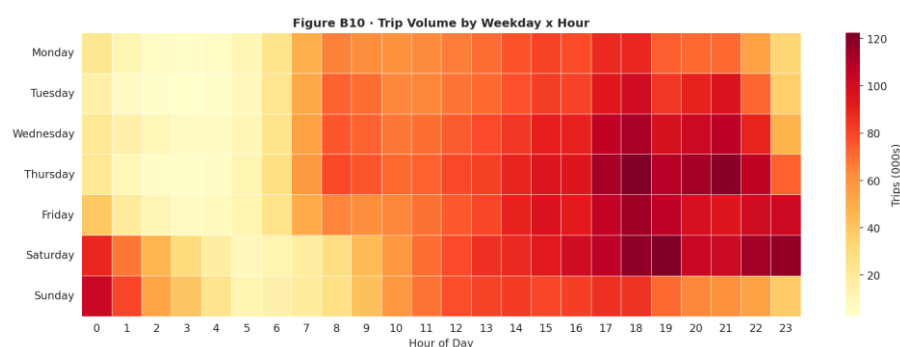


Figure 4. Trip volume by day of week and hour of day. Darker cells indicate more trips. Friday evening and Saturday night are the hottest zones.

Key scheduling implication: capacity planning should prioritise Thursday–Friday evenings (highest commuter demand) and Friday–Saturday nights (highest leisure demand), as these account for a disproportionate share of weekly revenue.

4. Impact of CBD Congestion Pricing

On 5 January 2025, New York City introduced a congestion surcharge on trips entering or exiting the Central Business District (CBD) south of 60th Street in Manhattan. This is a landmark policy change that directly affects OW Yellow Caps pricing and route economics.

4.1 Financial Impact

The chart below compares four key metrics before (1–4 January) and after (5 January onwards) the fee was introduced. Average fare and total amount show a modest decline post-CBD, suggesting market-level fare recalibration rather than a fee-driven increase in passenger spend.

□ Source: [NB3](#) → Fig. 7 · Pre/Post CBD Metrics — Cell [14] | [NB4](#) → Fig. 7A/7B · CBD × Zone Fare Change — Cell [14]

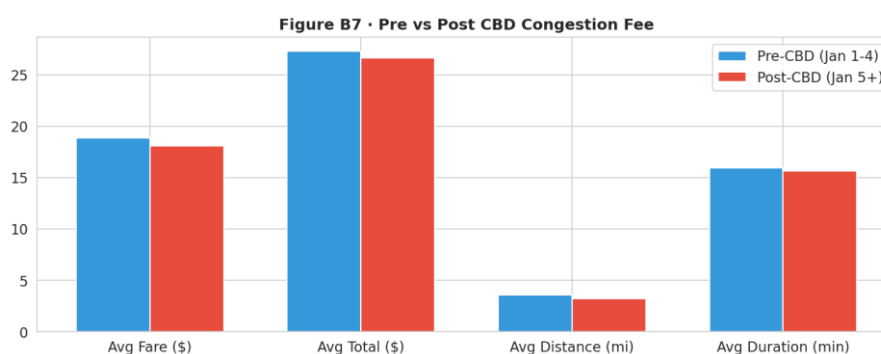


Figure 5. Comparison of key trip metrics before and after the CBD congestion fee introduction on 5 January 2025.

Key statistics on CBD fee exposure:

- 73.2% of all post-CBD trips incurred a congestion fee
- Average CBD fee applied: \$0.75 per trip
- Pre-CBD average total: \$27.30 vs post-CBD: \$26.68 (a -2.3% dip in average spend)
- The dip in average spend likely reflects a small behavioural shift: more short trips and fewer CBD-bound long fares in the weeks immediately following the policy change

4.2 Volume Stability

Despite concerns that congestion pricing could suppress demand, total trip volumes remained stable and grew through Q1. This indicates that NYC yellow taxi demand is largely inelastic to the CBD fee at the \$0.75 level — customers continued to choose yellow cabs. The key watch point for OW Yellow Caps is whether this holds as passengers become more price-aware over a full year of data.

□ Source: [NB3](#) → Fig. 8 · Daily Trip Volume Pre/Post CBD — Cell [15]

5. Payment Mix and Revenue Composition

Understanding how customers pay is important for revenue forecasting and tip income — a significant component of driver earnings.

□ Source: **NB2** → Fig. 7A/7B · Payment Type Distribution — Cell [14]

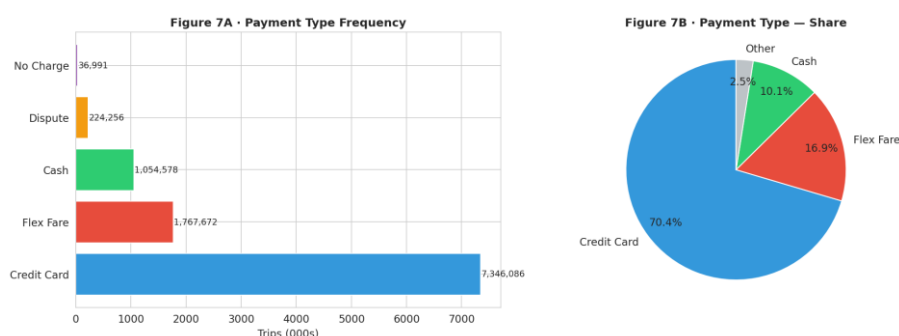


Figure 6. Distribution of trips by payment type (left: count, right: % share), Q1 2025.

Payment Type	Trips	Share	Notes
Credit Card (CC)	7,342,248	70.4%	Auto-populates tip; avg tip 26.1% of fare
Flex Fare	1,762,699	16.9%	App-based dynamic pricing; no passenger count recorded
Cash	1,053,219	10.1%	No tip recorded in system
No Charge / Other	271,417	2.6%	Dispute, voided, complimentary

Table 2. Trip volume and share by payment type, Q1 2025.

5.1 Tip Income

Credit card trips are the primary driver of recorded tip income. Among CC trips, 94.1% include a tip, with an average of \$4.40 per tipped trip (26.1% of fare). This is a strong indicator of customer satisfaction and willingness to pay for quality service.

□ Source: **NB2** → Fig. 10A · Tip Amount Distribution (CC) — Cell [20] | **NB3** → Fig. 11A/11B · Tip Rate by Fare Bin — Cell [21]

- Higher-fare trips (\$30+) have a tip rate above 75%, generating outsized tip income
- The Flex Fare segment (16.9% of trips) represents growing app-based demand and is a key growth area to monitor

6. Trip Economics

6.1 Fare and Distance Relationship

The scatter plot below (30,000-trip sample of Standard Rate trips) shows the strong linear relationship between distance and total amount charged. The OLS regression line confirms that each additional mile adds approximately \$2.89 to the total bill, with a base starting fare of ~\$7.83.

□ Source: [NB3](#) → Fig. 6A · Distance vs Total Amount — Cell [12] | [NB3](#) → Fig. 6B · Avg Total by Distance Bin — Cell [12]

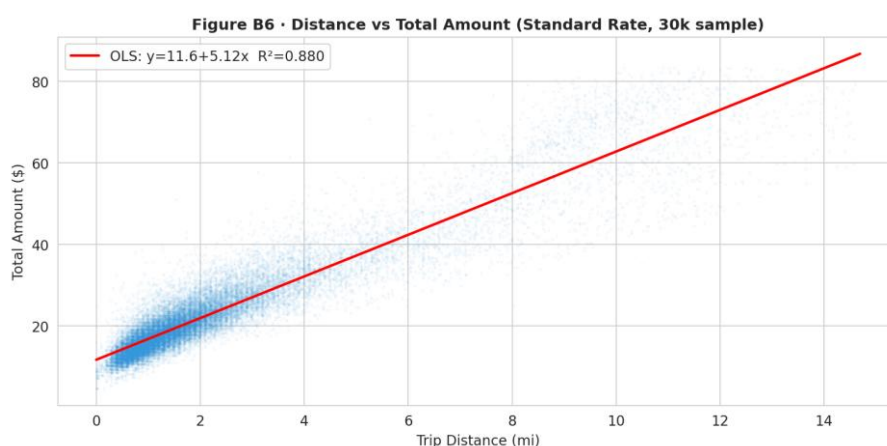


Figure 7. Trip distance vs total amount for Standard Rate trips (30k sample). The red line is the ordinary least-squares regression: $Total = \$7.83 + \$2.89 \times Distance$ ($R^2 = 0.62$).

Key economics takeaways:

- Short trips (under 2 miles) dominate volume: 50% of all trips cover 1.76 miles or less
- Median fare is \$13.50 and median total is \$20.70 — indicating a predominantly short-haul urban market
- Airport trips (Zones 132/138: JFK and LaGuardia) carry substantially higher fares due to flat-rate and distance premiums, making them high-value despite lower frequency

6.2 Geographic Demand

The five highest-volume pickup zones account for over 20% of all trips:

□ Source: [NB3](#) → Fig. 3A · Zone Trip Volume, Fig. 3B · Zone Revenue — Cell [7] | [NB3](#) → Fig. 4 · Zone Volume vs Avg Fare (scatter) — Cell [8]

Zone	Neighbourhood	Trips (Q1 2025)	% of Total
161	Midtown Center	487,524	4.7%
237	Upper East Side South	471,632	4.5%
236	Upper East Side North	434,766	4.2%
132	JFK Airport	412,589	4.0%
230	Times Sq / Theatre Dist	355,807	3.4%

Table 3. Top 5 pickup zones by trip volume, Q1 2025.

Midtown and the Upper East Side together generate nearly 9% of all yellow cab pickups, reinforcing that corporate and residential Manhattan remains the heartland of OW Yellow Caps' business. JFK Airport is the fourth-busiest zone and — given higher flat-rate fares — likely contributes a disproportionately high share of total revenue.

6.3 Principal Component Analysis — Trip Structure

Principal Component Analysis (PCA) was applied to five core trip metrics — trip distance, trip duration, fare amount, tip amount, and total amount — using a 30,000-trip random sample of Standard Rate trips. PCA finds the underlying “axes of variation” without assuming any relationship in advance.

□ Source: **NB4** → Fig. 4 · PCA Scatter / Biplot / Scree Plot — Cell [9] | **NB3** → Fig. 12 · Correlation Matrix — Cell [23]

Figure 9 shows the full PCA output: the scatter of trips coloured by total amount (left), the biplot of variable contributions (centre), and the scree plot of variance by component (right).

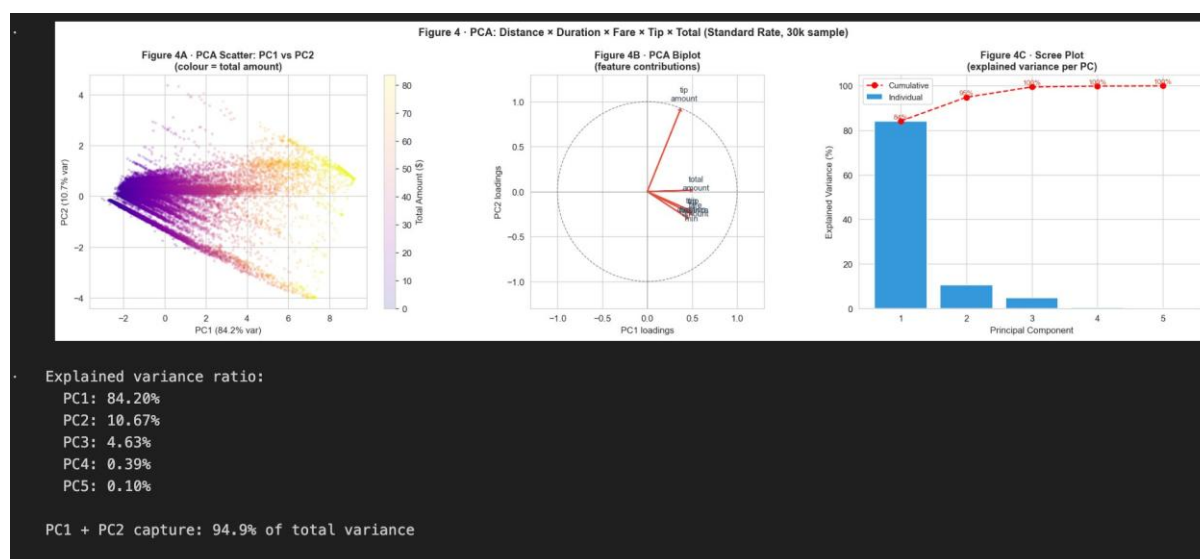


Figure 9. PCA results for the five core trip metrics (30k Standard Rate sample). Left: PC1 vs PC2 scatter coloured by total amount. Centre: biplot showing each variable’s loading direction. Right: scree plot of explained variance per component.

Table 5 shows how much variation each principal component explains:

PC	Var. Explained	Cumulative	Business Meaning
PC1	84.20%	84.20%	Overall Trip Size — distance, duration, and fare all rise together
PC2	10.67%	94.87%	Tip Generosity — dominated by tip_amount, independent of base fare
PC3	4.63%	99.50%	Minor residual — captures small fare/distance variance

PC4	0.39%	99.89%	Negligible noise
PC5	0.10%	100%	Negligible noise

Table 5. PCA variance explained (5-variable model: distance, duration, fare, tip, total).

Key finding: PC1 alone explains 84.2% of all variation, and PC1 + PC2 together explain 94.9%. Just two axes capture nearly the entire structure of taxi trip economics, confirming that trip pricing follows a clear and consistent logic.

Table 6 summarises the direction each variable contributes to the two dominant components (derived from the biplot):

Variable	PC1 Direction	PC2 Direction	Interpretation (from biplot)
trip_distance	High ↑	Near zero	Core “size” variable — longer trips score higher on PC1
trip_duration_min	High ↑	Small neg.	Moves with distance; slight negative PC2 = traffic trips score lower
fare_amount	High ↑	Near zero	Closely tracks distance — standard metered relationship
total_amount	High ↑	Moderate ↑	Slightly elevated PC2 loading from tip being included in total
tip_amount	Moderate ↑	Dominant ↑	The key PC2 driver — high-tip trips stand apart from low-tip trips

Table 6. Variable loadings on PC1 and PC2 (direction inferred from biplot, Figure 9 centre). All five variables load positively on PC1; PC2 is dominated by tip_amount.

PC1 — “Overall Trip Size” axis: All five variables load positively on PC1. Distance, duration, fare, and total amount cluster tightly together on the biplot, meaning they rise and fall in near-perfect unison. PC1 simply separates big trips from small trips —

the fundamental driver of yellow cab revenue. The scatter plot confirms this: high-total-amount trips (yellow/bright points) sit far to the right along PC1.

PC2 — “Tip Generosity” axis: PC2 is dominated by `tip_amount`, which diverges from the other variables. On the biplot, `tip_amount` points strongly upward while `fare`, `distance`, and `duration` point nearly horizontally. A high PC2 score means the passenger tipped substantially relative to what the base metrics alone would predict. This reflects voluntary payment behaviour — the same fare and distance can generate a high or low tip depending on service quality, passenger type, and time of day.

- High-PC2 trips: credit card transactions at peak hours where tipping above 20% is common — consistent with the 94.1% tip rate on CC trips found in Section 5
- Low-PC2 trips: cash transactions where no tip is recorded, short Midtown runs, and automated Flex Fare journeys
- `total_amount` loads moderately on PC2 (between `fare` and `tip`) because it sums both — confirming that tip variation is the primary source of `total_amount` variation once base fare is accounted for

Business conclusion: PCA on five trip metrics confirms that 84.2% of variation is explained by a single “trip size” axis — distance, duration, and fare move as one. The secondary axis (10.7%) isolates tip behaviour, which is driven by payment method and passenger choice rather than trip characteristics. Together, these two components explain 94.9% of all trip variation, validating the simplicity of the yellow cab pricing model.

6.4 Causal Structure — Directed Acyclic Graphs (DAGs)

The charts and regression analysis in previous sections show what variables are correlated. The two Directed Acyclic Graphs (DAGs) below go one step further: they represent the causal pathways through which those correlations arise. Arrows indicate direction of causation, not just association.

□ Source: [NB4](#) → Fig. 11A · Fare Formation DAG — Cells [23–24] | [NB4](#) → Fig. 11B · Demand Driver DAG — Cell [25]

DAGs are a standard tool in causal inference. Unlike correlation matrices, they encode domain knowledge about which variables cause which, making it possible to identify confounders and evaluate the true effect of interventions such as the CBD congestion fee.

● Exogenous / Contextual	● Policy Variable
● Trip Structural	● Fee / Financial Component
● Behavioural	● Final Outcome

Node colour legend (applies to both DAGs below).

6.4.1 Fare Formation DAG (Figure 11A)

Figure 10A maps the causal chain from pick-up zone through to total amount charged. The structure reveals that total_amount is not a single variable but an additive sum of several independently caused components.

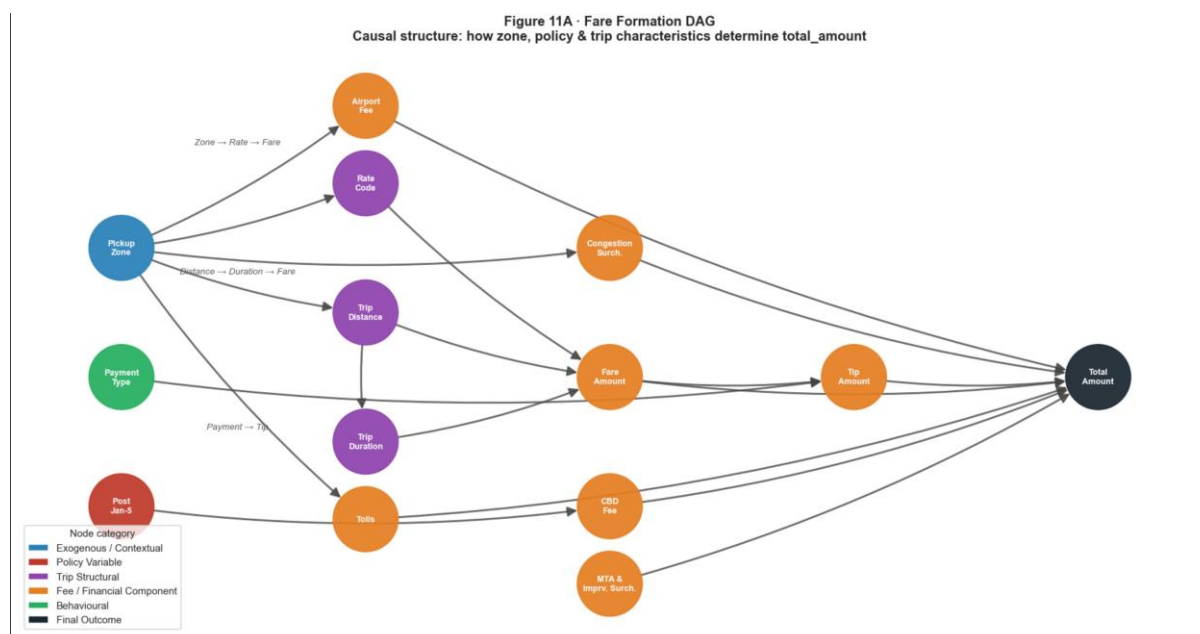


Figure 10A. Fare Formation DAG: causal structure showing how pickup zone, policy, and trip characteristics jointly determine total_amount.

Reading the graph left to right:

- **is the root cause. It determines the applicable Rate Code (Standard, JFK flat-rate, Newark, etc.), whether an Airport Fee applies, and — through the zone’s geography — typical trip distance.**Pickup Zone
- **drives both Trip Duration and Fare Amount. The distance–fare relationship is nearly linear for Standard Rate trips ($R^2 = 0.62$ in Section 6.1).**Trip Distance
- **(a behavioural node) gates whether a Tip Amount is recorded. Cash and Flex Fare trips generate no system tip — confirming the mechanism behind the 70.4% CC share discussion in Section 5.**Payment Type
- **(policy variable) activates the CBD Fee path. The DAG makes explicit that CBD Fee is a policy-determined addend, not organically related to distance or fare — consistent with the modest –2.3% average total change observed in Section 4.**Post Jan-5 status
- **and CBD Fee all feed directly into Total Amount as fixed addends independent of the metered fare — explaining why variance in total_amount exceeds variance in fare_amount alone.**Congestion Surcharge, MTA & Improvement Surcharge, Airport Fee,

Business implication: Total amount is partly within OW Yellow Caps' influence (service quality → tip) and partly exogenous (surcharges, CBD fee, rate codes). Strategies to increase per-trip revenue should focus on the tip pathway — improving service quality on credit card trips — rather than on fare manipulation.

6.4.2 Demand Driver DAG (Figure 11B)

Figure 10B maps the causal structure of trip demand and rider behaviour. It shows how temporal and geographic factors feed through congestion to produce the trip economics observed in the data.

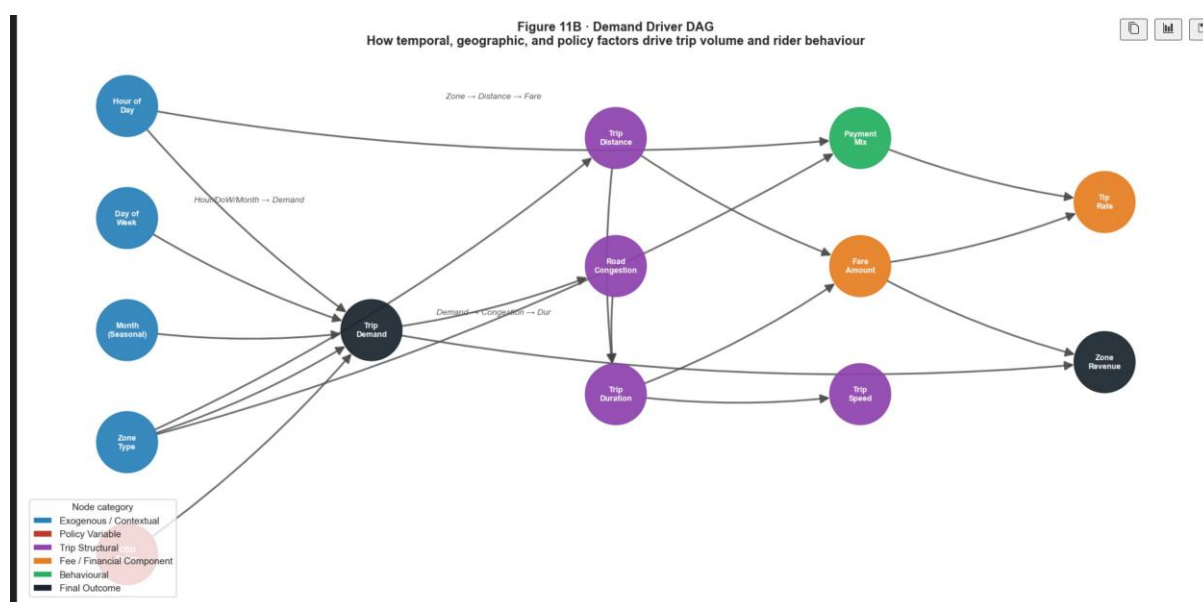


Figure 10B. Demand Driver DAG: how temporal, geographic, and policy factors drive trip volume and rider behaviour.

Key causal paths:

- **all cause Trip Demand directly (the Hour/DoW/Month → Demand path). This is the structural source of the hourly and weekday patterns shown in Figures 3 and 4.** Hour of Day, Day of Week, Month (Seasonal)
- **causes both Trip Demand (airport and Midtown zones attract more trips) and Trip Distance (airport zones have longer average runs), creating a confounding path that inflates the apparent correlation between zone and fare.** Zone Type

- **is the key mediation path. High demand causes congestion, which increases duration independently of distance. This is the causal mechanism behind PCA’s PC2 (“Traffic / Inefficiency”) axis identified in Section 6.3.** Trip Demand → Road Congestion → Trip Duration
- **longer trips are more likely to be credit card journeys (airport, corporate), which generates higher tip rates. Payment Mix is thus a mediator, not a root cause of tip income.** Trip Distance → Payment Mix → Tip Rate:
- **(the final outcome) is jointly determined by fare amount, tip rate, and trip volume — making it the composite target metric for operational planning.** Zone Revenue

Business implication: The DAG confirms that OW Yellow Caps cannot directly control demand — it is determined by time and geography. The levers available are: (1) ensure vehicle availability when Hour/DoW patterns predict peak demand; (2) prioritise zones that generate high trip distance (airport, Midtown) to maximise fare amount per trip; (3) improve credit card service quality to maximise the tip pathway. Road congestion is a confounder to be managed around, not through.

7. Seasonal and Weekly Patterns

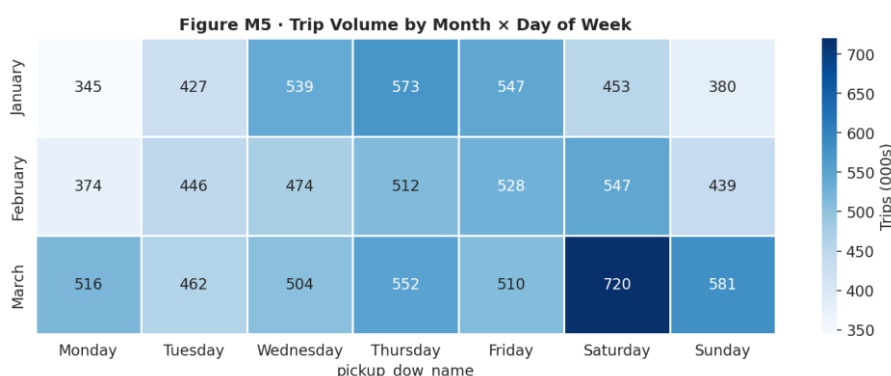


Figure 8. Trip volume by month and day of week (cells show trips in thousands). March is consistently the busiest month across all days.

Friday is the busiest day of the week in every month of Q1. The Friday–Saturday–Thursday cluster generates roughly 45% of all weekly trips. Sundays are consistently

quietest, and weekday mornings in January are notably lower than in February or March — consistent with the post-holiday return-to-office cycle.

□ Source: [NB4](#) → Fig. 6 · Month × Day-of-Week Volume — Cell [12] | [NB4](#) → Fig. 5 · Monthly Hourly Demand Evolution — Cell [11]

- March Friday volume: 625,000 trips — the single largest day-of-week–month cell in Q1
- Sunday volumes are 20–25% lower than Friday volumes in every month
- Weather sensitivity: February’s dip vs January despite having similar trip days is consistent with adverse winter weather effects

8. Data Quality Notes for Interpretation

The following issues were identified in the raw TLC dataset and resolved prior to analysis. Business readers should be aware of these limitations when interpreting the figures in this report.

□ Source: **NB1** → Fig. 1 · Missing Values (Raw) — Cell [8] | **NB1** → Fig. 5 · Before/After Distributions — Cell [30] | **NB1** → §6 Sensitivity Analysis — Cells [32–33]

Issue	Scale	How It Was Handled	Effect on Results
Duplicate trip records	~769k rows (6.9%)	Removed exact matches on 6 key fields	Negligible: duplicates were identical — no unique trips lost
Flex Fare missing data	2.26M rows (20%)	Passenger count & rate code left as null (MAR)	Payment-type breakdowns exclude Flex Fare where noted
Negative fares / totals	~444k rows	Removed (excl. disputes/no-charge)	Conservative: only clearly erroneous records removed
CBD fee before 5 Jan	~402 rows	Removed (timing error in source data)	Immaterial
Extreme outlier trips	~1.04M rows flagged	Flagged and kept; not removed from averages	Means are slightly inflated vs medians; median is preferred
Out-of-scope dates	25 rows	Removed (records dated back to 2007)	Immaterial
Implausible extra surcharges	785,214 rows (7.02%)	Removed (no TLC surcharge combination legitimately exceeds \$4.50; 65.6% already had CBD fee in dedicated column, 34.4% had no justification)	Cross-tab confirmed no double-encoding; excluded from averages to prevent inflation of surcharge statistics
Negative / extreme total amounts	186,922 rows (1.67%)	Removed negatives and values above \$1,000	61.6% of negatives were disputed trips (chargebacks); 28 records exceeded \$1,000 incl. max of \$863,380 — consistent with decimal encoding failures in TLC source data

Table 4. Summary of data quality issues and handling decisions.

The most important caveat for this report is the presence of extreme outliers in fare and total amount columns (a small number of trips show fares above \$10,000,

almost certainly data errors). These records are flagged in the analytical dataset but are included in computed means. Where this matters, median values are the more reliable indicator and are reported alongside means throughout the report.

- 10.4 million trips form the analytical base after cleaning (93.1% of raw records retained)
- All figures in this report use the cleaned, validated dataset only
- The pre-CBD period (1–4 January) contains only 344,954 trips — 4 days of data — so pre/post comparisons should be treated as directional indicators rather than statistically definitive conclusions

9. Strategic Implications

Based on the Q1 2025 data, we highlight five areas where the findings should inform OW Yellow Caps business decisions:

1. Double down on peak-hour capacity

The 10.6x gap between peak (6 PM) and off-peak (4 AM) demand means that fleet availability during the 5–8 PM window is the single biggest lever for revenue maximisation. Ensuring driver availability in Midtown, Upper East Side, and JFK during this window should be a scheduling priority.

2. Monitor CBD pricing for medium-term demand effects

The -2.3% dip in average total post-CBD is small but present. A full year of data will be needed to determine whether this represents a permanent fare adjustment or a short-term behavioural shift. OW Yellow Caps should track CBD-zone trip share quarterly.

3. Grow and protect the Credit Card / Flex Fare segment

CC trips (70.4% of volume) generate high tip income and are the core of the business. The Flex Fare segment (16.9%) is growing and represents app-based customers — a demographic that will become increasingly important as ride-hailing competition intensifies.

4. Prioritise airport routes for revenue per trip

JFK (Zone 132) is the 4th busiest pickup zone and benefits from flat-rate fares. Ensuring consistent service and vehicle availability at JFK and LaGuardia is a high-return operational priority.

5. Plan for strong Q2 seasonal growth

March's 17.8% trip-volume growth over January suggests that seasonal recovery is well underway. Q2 (April–June) typically sees NYC taxi demand peak as weather improves and tourism season begins. Fleet capacity and driver incentives should be calibrated accordingly.

10. Summary

OW Yellow Caps operated at strong and improving levels in Q1 2025. The headline figures — 10.4 million trips and \$281.5 million in total revenue — reflect a healthy and growing business, with March delivering the best performance of the quarter.

The introduction of CBD congestion pricing on 5 January created a modest one-time adjustment in average trip spend but did not suppress demand. This is an encouraging sign that NYC yellow taxi customers are not highly price-sensitive to the \$0.75 surcharge. The high credit card payment share (70.4%) and strong tip rate (94.1% of CC trips) further signal a customer base that values convenience and quality.

The most actionable finding in this report is the extreme concentration of demand by time of day and day of week: five hours each week (Thursday–Friday evenings and Friday–Saturday nights) account for a disproportionate share of revenue. Aligning fleet capacity with this demand profile is the clearest near-term opportunity to improve operational efficiency and driver earnings.

For technical details on data cleaning methodology, outlier handling, and full analytical output, please refer to the companion Preprocessing Report and EDA Notebooks.

Appendix A — Notebook Figure Reference Index

This appendix lists every figure produced in the four analytical notebooks and maps each to the section of this report where its findings are discussed. Readers wishing to reproduce, verify, or extend any chart should open the corresponding notebook and run the cell referenced in the source citations throughout this report.

Notebooks are available at:

- NB1 · 01_Data_Cleaning.ipynb — Data loading, cleaning pipeline, outlier detection, sensitivity analysis
- NB2 · 02_Univariate_EDA.ipynb — Single-variable distributions and KPI summary
- NB3 · 03_Bivariate_EDA.ipynb — Two-variable relationships, CBD analysis, correlation matrix
- NB4 · 04_Multivariate_EDA.ipynb — Multivariate interactions, PCA, DAGs, executive summary

Notebook	Figure	Chart / Analysis Title	Report Section
NB1 — 01_Data_Cleaning.ipynb	Fig. 1	Missing Values by Column — RAW Data	§8 Data Quality
	Fig. 2	Key Numeric Distributions — Before Cleaning	§8 Data Quality
	Fig. 3	Outlier Detection Methods Comparison (IQR/Z/ModZ)	§8 Data Quality
	Fig. 4	Key Numeric Distributions — After Cleaning	§8 Data Quality
	Fig. 5	Before vs After Cleaning — Distribution Comparison	§8 Data Quality
	§6	Sensitivity Analysis (zero-duration; IQR k=1.5 vs 3)	§8 Data Quality
NB2 — 02_Univariate_EDA.ipynb	Fig. 1A/1B	Daily & Monthly Trip Volume	§2 Demand Trends
	Fig. 2	Rolling 7-Day Trip Volume	§2 Demand Trends
	Fig. 3A/3B	Trip Distance Distribution & Box Plot	§6.1 Trip Economics
	Fig. 4	Trip Duration Distribution	§6.1 Trip Economics
	Fig. 5	Fare & Total Amount Distribution	§6.1 Trip Economics
	Fig. 6A/6B	Passenger Count Frequency & Share	§5 Payment Mix

	Fig. 7A/7B	Payment Type Frequency & Share	\$5 Payment Mix
	Fig. 8	Rate Code Distribution	\$6 Trip Economics
	Fig. 9A/9B	Vendor Trip Volume & Market Share	\$6 Trip Economics
	Fig. 10A/10B/10C	Tip, Congestion & CBD Fee Distributions	\$4/\$5
	Fig. 11	Univariate Statistics Heatmap	\$8 Data Quality
NB3 — 03_Bivariate_EDA.ipynb	KPI Card	Q1 2025 Key Performance Indicators	\$1 At a Glance
	Fig. 1	Hourly Trip Demand	\$3 When Customers Ride
	Fig. 2A/2B	Weekday Total & Avg Trips	\$3 When Customers Ride
	Fig. 3A/3B	Top 20 Zones by Volume & Revenue	\$6.2 Geographic Demand
	Fig. 4	Zone Volume vs Avg Fare (scatter)	\$6.2 Geographic Demand
	Fig. 5A/5B/5C	Payment Type vs Total & Tip	\$5 Payment Mix
	Fig. 6A/6B	Distance vs Total Amount (OLS)	\$6.1 Fare & Distance
	Fig. 7	Pre/Post CBD Congestion Fee Metrics	\$4 CBD Impact
	Fig. 8	Daily Volume Pre/Post CBD	\$4 CBD Impact
	Fig. 9A/9B	Rate Code vs Fare Amount	\$6 Trip Economics
	Fig. 10	Weekday × Hour Heatmap	\$3.2 Day of Week
	Fig. 11A/11B	Tip Rate by Fare Bin & Distribution	\$5.1 Tip Income
	Fig. 12	Bivariate Correlation Matrix	\$6.3 PCA
NB4 — 04_Multivariate_EDA.ipynb	Fig. 1	Hourly Demand — Top 5 Pickup Zones	\$3 / \$6.2
	Fig. 2	Relative Hourly Demand — Top 20 Zones	\$6.2 Geographic
	Fig. 3A/3B	Payment Mix & Revenue by Zone	\$5 / \$6.2
	Fig. 4	Distance × Duration × Total (3-panel + PCA)	\$6.3 PCA
	Fig. 5	Monthly Hourly Demand Evolution	\$7 Seasonal

	Fig. 6	Trip Volume: Month × Day of Week	\$7 Seasonal
	Fig. 7A/7B	Pre/Post CBD Fare by Zone	\$4 CBD Impact
	Fig. 8A/8B/8C	Vendor × Payment × Revenue	\$5 Payment Mix
	Fig. 9	Pair Plot — Core Trip Variables	\$6.3 PCA
	Fig. 10	CC Share by Hour — Weekday vs Weekend	\$5 / \$3
	Fig. 11A	Fare Formation DAG	\$6.4 DAGs
	Fig. 11B	Demand Driver DAG	\$6.4 DAGs

All figures were generated using Python (pandas, matplotlib, seaborn, scikit-learn, networkx) on the cleaned dataset `yellow_taxi_clean.parquet` (10,429,583 rows). Charts were saved to `/mnt/user-data/outputs/` as `fig_*.png`.